

Comparative analysis of the performance of RNN- and CNN-based generative models for the generation of synthetic location trajectories

SHYAM KAUSHIK, Indian Institute of Technology Delhi, India

ERIK BUCHHOLZ, University of New South Wales CSIRO's Data61, Cyber Security CRC, Australia

SALIL S. KANHERE, University of New South Wales, Australia

The generation of synthetic location trajectories is vital for applications in location-based services, privacy-preserving data analysis, and movement pattern simulations. This research proposes a comparative analysis of Generative Adversarial Networks (GANs) utilizing Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) to generate these trajectories. Specifically, we will examine LSTM-based TrajGAN and CNN-based models. LSTM TrajGANs, adept at handling sequential data and CNN-based models, excelling in feature extraction, present unique advantages and challenges in modelling spatiotemporal data. Utilizing a comprehensive dataset of real-world location trajectories, we will develop and train both models to generate synthetic trajectories. The performance of these models will be evaluated through a set of metrics that measure the quality, realism, and utility of the generated trajectories. This study aims to uncover the strengths and weaknesses of each model type in producing realistic and functional synthetic data. The results are expected to offer valuable insights into the effectiveness of LSTM and CNN architectures in trajectory generation, advancing the field of synthetic data creation.

Additional Key Words and Phrases: Synthetic Data Generation, Location Trajectories, Spatiotemporal Data, Generative Adversarial Networks (GANs), LSTM TrajGAN, Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Deep Learning, Location Privacy.

ACM Reference Format:

Shyam Kaushik, Erik Buchholz, and Salil S. Kanhere. 2024. Comparative analysis of the performance of RNN- and CNN-based generative models for the generation of synthetic location trajectories. 1, 1 (May 2024), 4 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 INTRODUCTION

Location trajectories are invaluable for various applications, from urban planning to transportation management. However, the release of raw trajectory data raises privacy concerns due to the potential disclosure of sensitive information. While traditional protection mechanisms, such as Differential Privacy (DP), offer some level of privacy, they often compromise utility and are susceptible to attacks. Synthetic trajectory data generation presents a promising alternative, offering both privacy and utility benefits. Generative models have emerged as powerful tools for generating synthetic data that closely resemble real trajectories. In particular, Recurrent Neural Network (RNN)-based generative models, such as LSTM TrajGAN [4], and Convolutional Neural Network (CNN)-based models have shown potential in capturing the spatiotemporal characteristics of trajectories [2, 5]. This research proposal aims to conduct a comparative analysis of RNN- and CNN-based generative models for the generation of synthetic location trajectories. Specifically, we will explore the performance of LSTM TrajGAN and CNN-based models in terms of trajectory realism, accuracy, and utility. By evaluating these models using appropriate

Authors' Contact Information: Shyam Kaushik, ch221447@iitd.ac.in, Indian Institute of Technology Delhi, New Delhi, India; Erik Buchholz, University of New South Wales and CSIRO's Data61, Cyber Security CRC, Sydney, NSW, Australia, e.buchholz@unsw.edu.au; Salil S. Kanhere, University of New South Wales, Sydney, NSW, Australia, salil.kanhere@unsw.edu.au.

© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.

This is the author's version of the work. It is posted here for your personal use. Not for redistribution. The definitive Version of Record was published in , <https://doi.org/10.1145/nnnnnnn.nnnnnnn>.

metrics, we aim to provide practical guidance for researchers and practitioners seeking to generate realistic and privacy-preserving synthetic trajectory data [2].

2 BACKGROUND

This section discusses the background knowledge for this work. Section 2.1 formally defines trajectory datasets and introduces the datasets used for evaluation. Section 2.2 provides an introduction about the generative models.

2.1 Trajectory Datasets

A trajectory dataset D comprises multiple trajectories, denoted as $D = \{T_{u1}, T_{u2}, \dots, T_{zm}\}$, where T_{ui} refers to the i^{th} trajectory of user u . Each trajectory T is represented as an ordered sequence of locations: $T = (l_1, \dots, l_n)$, where each location $l_i = (lat_i, lon_i)$ typically includes latitude and longitude coordinates. Trajectories may also contain semantic information such as Points of Interests (POIs) (e.g., restaurant, shop, gym) to provide context. While semantic information enhances dataset utility, facilitating tasks like Trajectory User Linking (TUL), our focus is on trajectories with spatial coordinates, as not all datasets include semantic details.

2.2 Generative Models

Convolutional Neural Networks (CNNs) are a class of neural networks particularly well-suited for processing spatial data such as images. Unlike traditional fully connected networks, CNNs use a hierarchical structure of convolutional layers to automatically learn & extract spatial features from input data. In the context of synthetic trajectory generation, CNNs offer valuable capabilities in extracting spatial features from trajectory data.

Recurrent Neural Networks (RNNs) represent a specialized neural architecture tailored for processing sequential data. Unlike traditional feedforward networks, RNNs feature connections forming directed cycles, enabling them to retain memory of previous inputs. This attribute renders them ideal for tasks involving time-series data analysis, including natural language processing and trajectory prediction.

Generative Adversarial Networks (GANs) are a pioneering class of machine learning models introduced by Goodfellow et al. [3]. Comprising two neural networks – a generator and a discriminator – GANs are trained in tandem within a competitive framework. The generator aims to produce synthetic data resembling real samples, while the discriminator distinguishes between genuine and artificial data.

Autoencoder (AE) AEs [1] consist of an encoder that encodes a given input into a lower-dimensional latent representation and a decoder that aims at reconstructing the original input. Common applications are dimensionality reduction and denoising. The latent bottleneck forces outputs to be different, while the most important information is preserved if the model is trained correctly.

3 GOALS

The primary goals of this research project are outlined below, detailing the specific objectives and expected outcomes that will guide the study from proposal documentation to the dissemination of findings.

G1: Research Proposal Documentation

Formulate and document a comprehensive research proposal outlining the objectives, methodology, and expected outcomes of the project.

G2: Development of TrajGAN Models

Successfully develop CNN-based TrajGAN model for generating synthetic location trajectories.

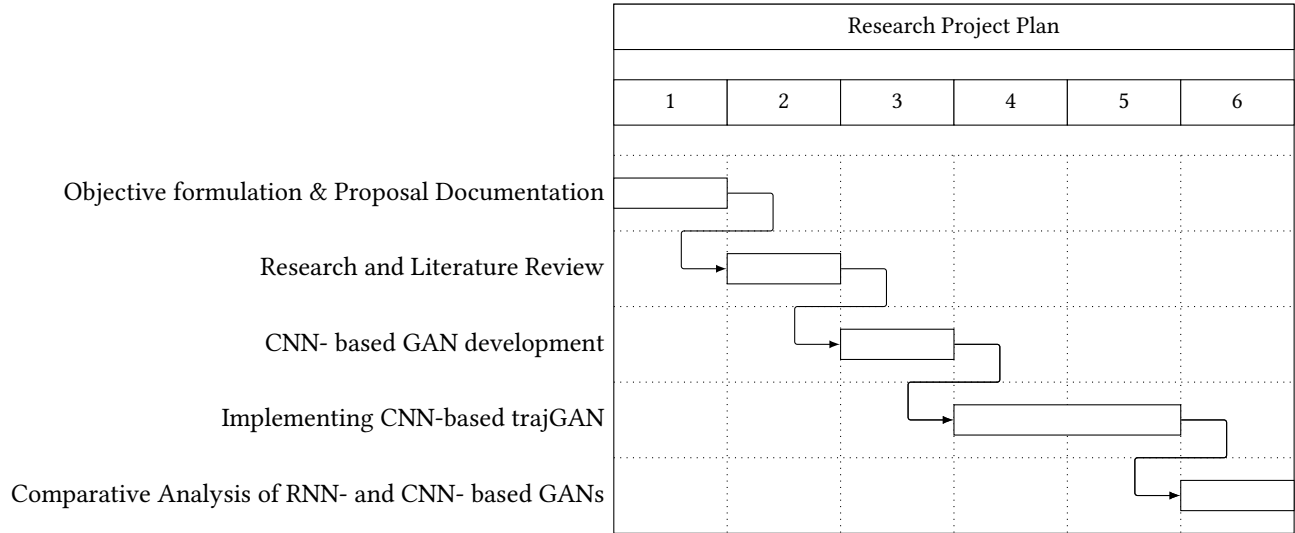
G3: Performance Evaluation Framework

Establish a robust framework for evaluating the performance of generative models using relevant metrics.

G4: Comparative Analysis

Conduct a comprehensive comparative analysis of RNN- and CNN-based TrajGAN models to identify their strengths and weaknesses in generating synthetic location trajectories.

4 TIMELINE



4.1 Week 1: 20 May - 26 May

Define project objectives and document them in the proposal. This includes outlining research questions and methodology

4.2 Week 2: 27 May - 2 June

Conduct a comprehensive literature review to identify relevant research articles and publications. This involves understanding existing work on generative modelling and trajectory data analysis.

4.3 Week 3: 2 June - 9 June

Develop the architecture of CNN-based Generative Adversarial Networks (GANs). Begin model training using trajectory data and tune hyperparameters.

4.4 Week 4&5 : 9 June - 23 June

Implement the CNN-based Trajectory Generative Adversarial Network (TrajGAN) framework. Utilize preprocessed trajectory data, integrate the CNN-based GAN model, and optimize parameters for performance.

4.5 Week 6: 23 June - 30 June

Evaluate the performance of both RNN-based and CNN-based TrajGAN models. Compare their ability to generate realistic and diverse synthetic location trajectories

5 CONCLUSION

This research aims to perform a comparative analysis of RNN- and CNN-based generative models, specifically focusing on their performance in generating synthetic location trajectories. By developing and evaluating the TrajGAN models, this study will determine the efficacy of each approach in capturing the spatiotemporal patterns of real-world trajectories. The expected outcomes include a detailed performance assessment based on predefined

metrics, highlighting the strengths and limitations of each model. This research will contribute to the field by providing a framework for selecting appropriate generative models for synthetic trajectory generation, ultimately enhancing the utility and privacy of trajectory data in various applications.

REFERENCES

- [1] Noam Koenigstein Dor Bank and Raja Giryes. 2021. Autoencoders. *arXiv* (2021). <http://arxiv.org/abs/2003.05991>
- [2] Shuo Wang Surya Nepal Erik Buchholz, Alsharif Abuadbba and Salil S. Kanhere. 2024. SoK: Can Trajectory Generation Combine Privacy and Utility? *Proceedings on Privacy Enhancing Technologies* 2024, 3 (July 2024). <https://doi.org/10.48550/arXiv.2403.07218>
- [3] Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. 2014. Generative Adversarial Networks. *Commun. ACM* 63, 11 (2014), 139–144. <https://doi.org/10.1145/3422622>
- [4] Yuhao Kang Qunying Huang Jinmeng Rao, Song Gao. 2021. LSTM-TrajGAN: A Deep Learning Approach to Trajectory Privacy Protection. *Proceedings of the 11th International Conference on Geographic Information Science (GIScience 2021)* (2021), 16 pages, 7 figures. <https://doi.org/10.48550/arXiv.2006.10521> arXiv:arXiv:2006.10521 [cs.LG]
- [5] Xiang H Liang A Christensen HI Paz D, Zhang H. 2023. Conditional Generative Models for Dynamic Trajectory Generation and Urban Driving. *Sensors* 23, 15 (2023), 6764. <https://doi.org/10.3390/s23156764>

A GLOSSARY

CNN Convolutional Neural Network.

GAN Generative Adversarial Network.

RNN Recurrent Neural Network.

Received 24 May 2024; revised 12 March 2009; accepted 5 June 2009